

The Economy

Seminar 6: Markets and Market Failures

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Courtesy of
Zydney Wong

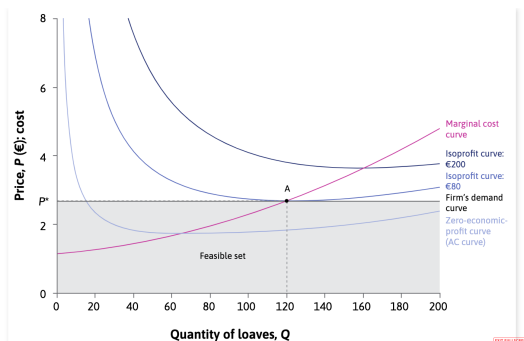
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Firms vs. Markets

- so far, we have focused on how one single firm interacts with workers on the labour market and with consumers on the goods market
 - when a firm is alone in the market, it has **market power** and are **price-setters**: they set a price (or choose a quantity corresponding to a price) that gives them profits
 - then the consumers are price-takers since they can choose the quantity they purchase given a price set by the firm
- in the other extreme, the economy is more of a market - we have a **large number of sellers and buyers** trading over **relatively homogeneous goods and services** in an **independent, uncoordinated way** and with **full access to all the prices of all the possible goods**
 - these conditions for a *perfect competitive market* hardly ever exist - most markets are somewhere between the two
 - in a perfectly competitive market, both firms and consumers are **price-takers**: they choose the quantity of good they want to sell/buy and the market (= the aggregation of all the buyers and sellers) determines the price
 - this is because many firms sell the same good, so any firm that tries to sell above the market price will get priced out (consumers can go elsewhere)
 - the flipside is that at this market price, a firm can sell as much as it produces since there are a lot of consumers, so they will never try to undercut below the market price to gain consumers

Firm Optimization in a Competitive Market

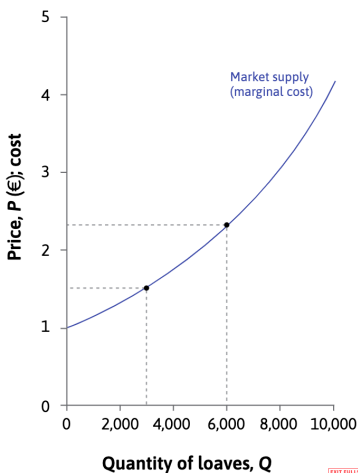
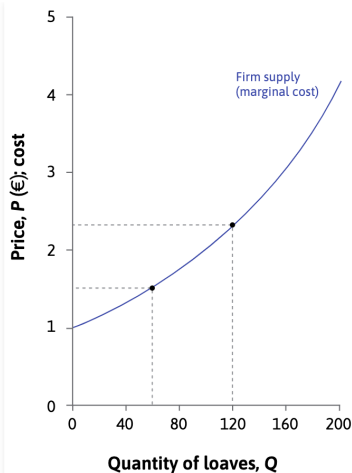


- this means for a **single firm**, the demand curve they see is horizontal (for all quantities, the price is the same)
 - given that the optimum is still where the isoprofit = the demand curve, firms produce at the minimum point of the isoprofit curve where slope = 0
 - recall that the marginal cost curve intersects the isoprofit curves at their minimums, so this is also where price = marginal cost
- the quantity and the price at the optimum for a **single firm** is such that price = marginal cost: $P = C_m(Q)$

Supply in a Competitive Market

- previously, a price-making firm chose $P > C_m(Q)$ because at this price level, selling an additional unit may earn them additional revenue but comes at the expense of losing profit on each of the previous units, which turns out to be a net loss in profit
- now this is not a problem, since firms are price-taking: firms need not decrease its price to attract more customers, so selling more no longer is subject to the price-quantity trade off
 $\Rightarrow$ so it sells until selling another unit costs too much and generates a loss:
 $P = C_m(Q)$
- this implies that the equilibrium quantities sold by a firm depend on the price of the market (set as an upper bound for the marginal cost) = **supply curve**
 - recall: total supply is the sum of the quantity produced (supplied) by each firm at a given price level
 - then the supply curve is the relationship between the price of a good and its total supply
 - quantity supplied increases with the price level = upwards sloping
 $\Rightarrow$ each firm can supply more when the price level increases because this raises the threshold of production at which price is outstripped by the marginal cost

Supply in a Competitive Market



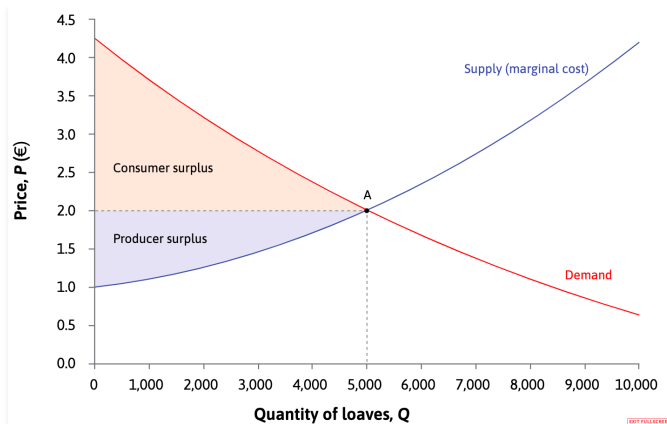
We aggregate firms' individual supply curves (left) to get an aggregate supply curve for the whole market (right).

EXIT FULLSCREEN

Demand in a Competitive Market

- here it is important to make a distinction between 3 types of demand:
 - ① the demand curve from consumers is **downwards sloping**
⇒ demand decreases as the price increases (since fewer people have sufficiently high WTP)
 - ② the demand curve addressed to each firm in a market is **flat**
⇒ each firm observes demand as a function of quantity it should produce given a certain price level; but since they are price-takers, the demand they observed cannot be adjusted to along the price dimension = flat horizontal line)
 - any deviation to charge a higher price will price them out of the market; any deviation to charge a lower price is unnecessary since they can sell to however many agents they want as long as it is profitable for them to do so
 - ③ the demand curve in the whole economy is **downwards sloping**
 - an individual firm may not change its price, but the market price may change
 - a higher market price leads to a lower demand from consumers

Competitive Equilibrium

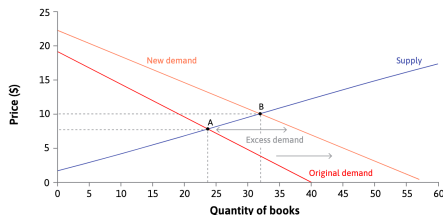


- A **perfectly competitive market equilibrium** is a (quantity, price)-pair such that
 - firms and consumers are price-takers who choose quantities they find optimal, given a prevailing market price
 - supply equals demand at this price (the market "clears")

Competitive Equilibrium

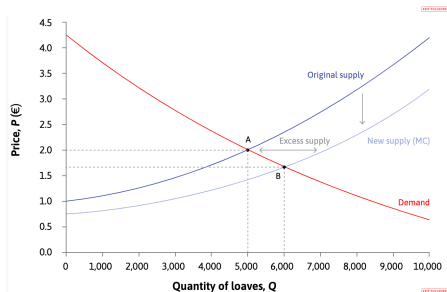
- This equilibrium is stable to disparities in quantities demanded and supplied as the price adjusts accordingly to go back to the equilibrium :
 - if for a given price, demand $>$ supply (i.e qty demanded is higher than the qty supplied), then firms can supply more to these willing buyers but would need a higher price to do so in order to offset the increased costs of production; they do so until supply = demand + no incentive to sell more
 - if for a given price, demand $<$ supply (i.e quantity demanded is lower than the quantity supplied), then there are not enough buyers for the supply that firms have produced, so firms produce less and only require a lower price to offset the costs of production; again, they do so until supply = demand
- The equilibrium will change if there are external (exogenous) non-market shifters in demand or supply:
 - demand shifters are changes in preferences, income or even the demand of other related goods (certain market events that make people have a higher willingness to pay or more people have a given willingness to pay)
 - supply shifters are changes in technology or costs of production or the supply of other related goods (e.g. the arrival of conveyor belt nanotechnology meant that we could produce nanomachines faster, so at each price, we can now have a higher supply)
- **note the difference between QUANTITIES demanded and supplied, and the actual DEMAND and SUPPLY itself**
 - a change in prices will always lead to shifts in qtys demanded and supplied
 - exogenous shifters move entire demand and supply curves + in reaction to a shift in these curves might be a reaction in the qtys supplied and demanded

Demand and Supply Shifters



- increased pref. for goods: a positive demand shifter

- demand curve moves from D to D' : qty demanded increases for each price
- at old price, there is now excess demand, so more buyers are willing to buy at this price
- supply curve is unchanged but the qty supplied increases to clear the market but need to do so with higher prices
- new eqm is at higher p + higher qty

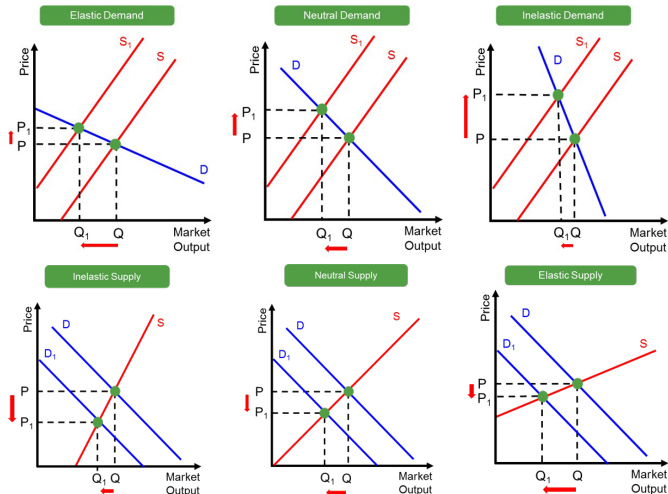


- technology improves: a positive supply shifter

- supply curve moves from S to S' : qty supplied increases for each price
- at old price, there is now excess supply, so firms want the price to fall
- demand curve is unchanged BUT qty demanded increases to clear the market and can do so with lower prices
- new eqm is at lower p + higher qty

NB: positive shifters of demand or supply will always increase eqm qty, but effect on price depends on which shifter

Demand and Supply Shifts with Elasticity

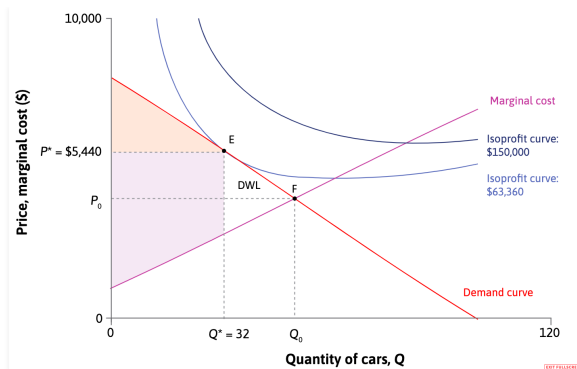


More elastic D/S $\rightarrow$ Q moves more than P does after S/D shift

Gains From Trade: Competitive Equilibrium

- same concept as previously: firms and consumers only trade if they benefit from it
 - firm surplus comes from the profits + fixed costs = total revenue - total marginal costs
 - consumer surplus comes from the sum of all the differences between WTP and actual price paid
 - social surplus is the sum of consumer surplus and firm surplus
- **at the competitive equilibrium, the total (social) surplus is maximised = no deadweight loss**
 - this is Pareto-efficient because it is impossible to produce more surplus + any re-allocation of this same surplus would mean reducing the surplus of any agent
⇒ (a higher price benefits the firms at the expense of the consumers; a lower price benefits the consumers at the expense of firms)
 - however, producer's surplus here is lower in competitive equilibrium than in the one with market power
⇒ this means that going from the market power equilibrium to a competitive equilibrium is not a Pareto improvement (even if it means getting a Pareto efficient outcomes)
 - which might explain why the GAFAM really struggle to let go of their market power and allow more firms to enter (it automatically reduces their firm surplus)

Summing Up...



You should be able to compare:

- how the price and quantity at the equilibrium are reached
 - if you're a price-taker (competitive market): cannot set prices yourself and choose quantity where $p = MC$, since producing more would lead to loss (cannot make profits with higher prices because other firms will undercut)
 - if you're a price-maker (single firm): choose the quantity to maximise your profits ($MR = MC$), which given the demand curve, allows firm to set price higher than MC (cannot increase quantities any more since marginal profit no longer offsets fall in price)
- effect on consumer surplus, producer surplus and deadweight loss (taking price-taking competitive market as the baseline) + Pareto efficiency

Market Failures: Market Power

- Recall that market power is in essence the **ability to set prices** (and quantities)
 - a monopolistic firm sets a quantity lower than the perfectly-competitive level, and a price higher than the perfectly-competitive price, so as to make a profit
 - this underproduction excludes some consumers who could have consumed from the market, and is thus socially suboptimal = **deadweight loss** $\Rightarrow$ monopolistic equilibria are not Pareto-efficient
- Market power exists because of the monopolistic position a firm holds in a market: to protect it, it must outshine potential entrants into the industry or stop the entrants for entering altogether
 - product differentiation at two stages: new or innovative product development (makes it hard for firms to copy and enter), advertising and marketing (to distinguish product from other competitors)
 - entry barriers: natural barriers (due to high sunk costs) or man-made barriers (due to industry regulations, cartels, patents etc.; high switch costs between service providers)

Natural Barriers to Entry

- natural monopolies in the provision of goods that have **high, fixed costs**:
 - it is socially optimal to have one (monopoly) or a few (oligopolistic) firms to invest in paying the high fixed cost; excessively replicating fixed costs is suboptimal
 - a consequence of the high fixed costs is that the AC curve takes a very long time to reach its minimum, and so the competitive equilibrium outcome where $P = MC$ tends to be quite a way away from the monopolist equilibrium where $MR = MC$
 - this is usually the case where substantial infrastructure is needed to deploy the service: railways, telecommunications, energy, water
 - it is also often individual suboptimal to have too many firms in a market with high fixed cost:
 - recall that the producer surplus is the profits + fixed costs; if fixed costs are high and the profit margin falls as the number of firms increase - the actual operating profit might be negative
 - this is the reason why the Big Three in French telecommunications have been so active in trying to exclude Free from the market
- while the **number of firms** (one or low) may be socially optimal, each firm still chooses a price that is too high and a quantity that is too low, and thus its **equilibrium actions** are socially suboptimal
 - there is always a tradeoff between too many firms (too much replication of fixed costs) and too few firms (equilibrium production too suboptimal): there is room for regulation

Natural Barriers to Entry

- any regulatory action could **separate the fixed cost** (of deploying and maintaining the network) **from the variable cost** (using the network to provide services)
 - this allows the high sale prices of services (due to the variable costs - which tends to be transferred on to consumers) to be kept low via competition
 - this is the case of railroads and electricity all over the world (except the UK where the introduction to competition was done to both network maintenance AND service provision simultaneously - and it was and is still a disaster)
 - but also think large F&B industrial groups that take the high factory sunk costs and run competitive franchises in nearly competitive markets
- alternatively, **directly regulate the price: set price = marginal cost**
 - keep the price low (so consumer welfare is protected) + subsidising the difference between AC and MC helps keep the costs manageable for firms (so producer surplus is protected)
 - subsidies is a double-edged sword: if your producer surplus is mainly subsidies, it means you have no incentive to product differentiate well and no incentive to innovate (to lower production costs, for instance)
 - also subsidies require a substantial and costly study of the market to choose the right level of optimal subsidy, and even then it is not always accurate (firms misreport profits to get higher subsidies)
- or, set price = average cost so firms never make a loss, but that means we need information on the true AC (same informational problems)
 - setting price = average cost is not always welfare improving (it depends on where you are along your AC curve)

The Digital Economy

- digital technology has **huge economics of scale**:
 - **server space** (both physically and in terms of bandwidth) is **incredibly cost-intensive**: the sunk costs are insanely high (which is why Amazon and Google's major new-industry breakthrough was renting out server space)
 - but once server spaces are up and running, marginal costs are close to zero
- so the industry has gradually become more and more **concentrated** in the hands of GAFA(M/N)
 - **network effects favour high-capacity firms**: more users generate more positive externalities on other users (improved algorithms for search engines, improved convenience with high user counts); more users means more externalities and a larger advantage
 - think about how the new Whatsapp privacy regulations was *supposed* to have a chilling effect on users, but in the end because no other platform has all your friends on it at once, no one wants to move
 - think about how it is easier to buy all of your random stuff on Amazon (a book + a Kindle + some cups) in one command then trying to do it across three commands from three smaller sites
 - early numerical firms made their services free and **financed their operations via advertising and data collection**
 - this means you have an increasing + convex marginal revenue curve and this means an increasing + convex marginal profit curve
 - and this is **effectively a barrier to entry** (for low-user firms on same technology, it's nearly impossible to get to a sufficiently high user count to get profit margins that rival the big firms)
 - the new tech breakthroughs of the last decade were all on radical new services/media: Spotify for streaming music (Deezer tried but failed to takeoff), Netflix for movie streaming (still going strong despite HBO+/ AmazonPrime/ Hulu/ Disney+), Tiktok (short videos), Snapchat (anonymity), Instagram (picture/video sharing)

Regulating The Digital Economy

- **regulating the digital economy cannot go through price regulation or controls:** this is not how these firms have market power
 - in fact, the ad/data market is quite competitive (across the big GAFAM/N)
- so regulation of the digital economy is notoriously difficult and requires **unorthodox regulation** that itself is difficult to implement:
 - lawsuits about data privacy laws against Youtube and Google = **reduce the scope of data that can be collected** makes these big firms less attractive (so lower profitability)
 - regulating algorithms is incredibly challenging because they are proprietary secrets and regulators have little to no access to the inner workings of these algorithms: but **reducing what type of information algorithms can use to learn** (so less data in its input, less field for its use) lowers how attractive they are
⇒ and this is ignoring all that has to do with consumer rights, privacy concerns (even if the desire to regulate begins here, but its impact is on the profitabilities of these companies)
 - **antitrust/anti-anti-competition measures** stop firms from agglomerating their user capacities (Facebook-Instagram interconnectivity, what you watch on Youtube changes what you see in ads on Google)
⇒ Facebook condemned for buying Instagram and Whatsapp = it almost singularly dominates the online communication market

⇒ Recent: [Apple antitrust case](#)



U.S. Sues Apple, Accusing It of Maintaining an iPhone Monopoly

Ian C. Bates for The New York Times



Artificial Barriers to Entry

- regulators (states) can set up artificial **barriers to entry** in some markets by :
 - requiring potential new firms to overcome costly administrative obstacles (authorisation to produce/sell, licences to operate)
 - blocking access to innovations that lower costs (patents, high costs for freedom of information requests)
- barriers to entry **prevent new firms from eroding market-dominant positions of incumbent firms = market power**
 - but this is not always a bad thing - at times we put in barriers to protect sensitive industries (fuel, food, urban transport)
 - lowering competition means firms focus less on widening consumer base and more on product innovation
 - think about how the original smartphone industry had a good ten companies fighting to promote almost identical phones, but as the number of firms fell, we had a substantial amount of product differentiation and innovation
 - empirically, the introduction/removal of barriers to entry tend to increase/decrease the profit margins of the incumbent firms
 - introducing barriers to entry makes it harder for smaller entrant firms to enter the market, and so allows firms to increase their profit margin (at the expense of consumers) = less risk of new entrants undercutting the market
 - removing barriers to entry (like in trucks) lowers costs for both incumbent and entrant firms, which leads to more truck transportation (more truckers employed)

Cooperative Market Power: Collusions and Cartels

- large firms can come together and **negotiate on a cartel-level price that is usually substantially higher than what they would do if they competed**
 - this interaction can be modelised game theoretically: firms choose to set a high price (cooperatively) or low price (competitively)
 - setting high prices together means you share more profits; setting low prices together means you share less profits; but setting a low price while the other sets a high price means you get all of the profits (you undercut the other), and this is usually more than the cartel-shared profits
 - **any form of cooperation between firms in markets is thus not a Nash equilibrium**: companies have incentive to deviate from high to low if the other company plays high
 - so cooperation can only be sustained either through the permanent retaliation in an infinitely repeated game (collusion) or via contractual obligations (cartels)
- here there is **nothing socially optimal about collusions and cartels**: they generate massive dead-weight losses at the expense of consumers
- usually they are illegal nationally and regulated by anti-trust laws and watchdog organisations, but with the porous economic legal framework internationally/transnationally, it is hard to stop countries from colluding
 - OPEC: oil cartel sets prices artificially (but cooperation is not a NE and hard to enforce)
 - collusions are also vulnerable to deviations if one firm can substantially benefit from whistle-blowing

The Empirics of Competition

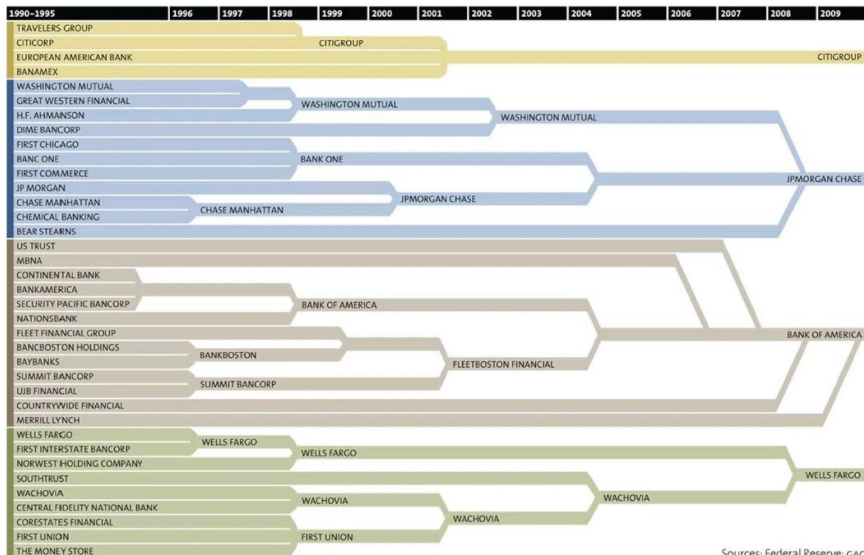
- the rise of concentration, cooperation and collusion across markets has meant that in general, market power has increased for a smaller number of firms, and **competition has decreased** in the last decades:
 - we see this by how large chunks of consumer markets (across household goods, services and technologies) are often concentrated in the hands of a few firms
 - the role of GAFAM in the tech industries: dominant positions and market shares, as of 2019-20

	Search engine	Operating systems	Social Media	Cloud	Ads	e-Commerce
Google	91	32	3	7	30	< 1
Apple	-	14	< 1	-	-	-
Facebook	-	-	70	-	22	-
Amazon	< 1	< 1	< 1	47	9	20
Microsoft	4	20	< 1	22	< 1	< 1

The Empirics of Competition

- this problem is particularly stark in the US: the average firm has become **three times larger** in the last 20 years (= fewer firms, more market power/firm)
- US firms have increased their profit margins and the degree of concentration = prices have increased by 15% in the US and wages by 7% in the last two decades + drive up inequalities (profit-driven dividends grow faster than wages)
 - 90% of the US beer market is owned by 2 firms
 - the **entire** beef market is owned by 4 firms
 - half the consumer banking sector in the US is basically 4 (or 5) firms (JPMorgan, Bank of America, Citigroup, Wells Fargo); in France (BNP Paribas, Crédit Agricole, Crédit Mutuel, Groupe BPCE (your neighbourhood *banque poulaires* and *caisse d'épargnes*), Société Générale)
- EU has done slightly better: tougher competition policy have helped protect customers
 - European firms don't just produce for Europe: they have a world market to deal with
 - global trade liberalisation post-WWII is another form of trade cooperation (or even collusion) on a global level and is subject to the same problems that any cooperation has in a game theoretical sense
 - Europe deciding to deliberalise (tougher competition policy, more anti-trust etc) means lower profits for firms unilaterally but US and China have no need/desire to follow - until recently

Banking Industries: US

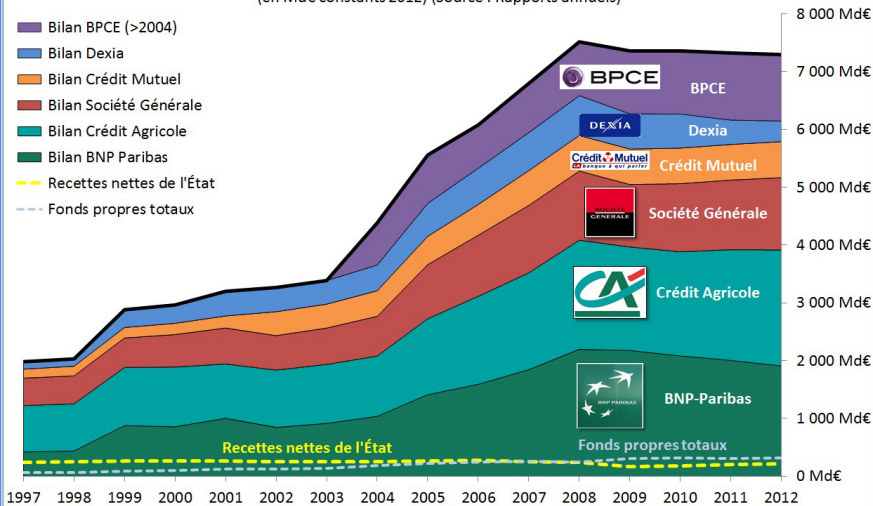


Sources: Federal Reserve; GAO

Banking Industries: France

Actif total des 6 premières banques françaises, 1997-2012

(en Md€ constants 2012) (Source : Rapports annuels)

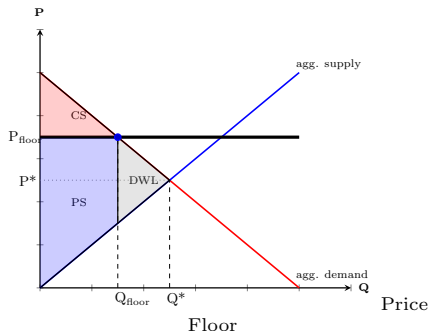


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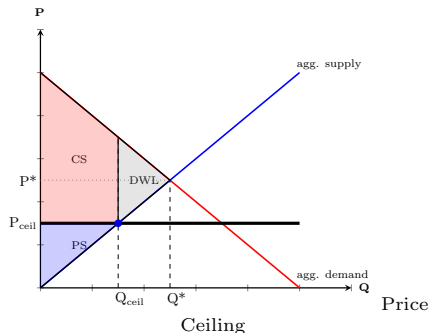
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Market Power: Price Distortions

- one way of regulating a market is to set price floors (minimum prices) or price ceilings (maximum prices) to protect vulnerable producers (risk-intense, waning industries usually) and vulnerable consumers (of bubble industries) respectively
 - the tradeoff is a surplus deadweight loss (= not Pareto efficient) and a redistribution of surplus from consumer to producer (price floors) or from producer to consumers (price ceilings) = so implementing price distortions is never a Pareto improvement



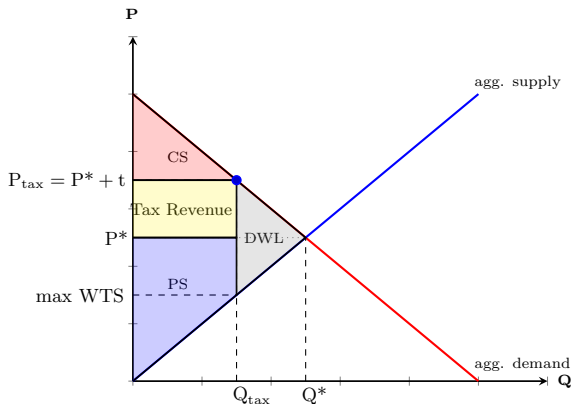
- price floor stops some consumers from consuming



- price ceiling stops some producers from selling

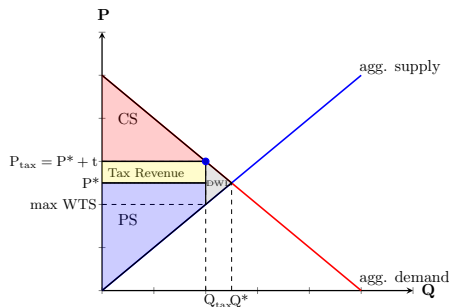
Market Failure: Taxation

- a sales tax t on a given good means that a consumer has to pay $p + t$, of which the producer receives p and a regulator (the state) receives t
 - this means total surplus is lower and also redistributed from consumers and producers to the regulator (so neither Pareto improvement from competitive equilibrium, nor Pareto efficient)
 - here: the lower equilibrium quantity prevents both some consumers from buying AND some producers from selling
 - this generates a tax wedge ($P_{tax} - \max \text{WTS}$)

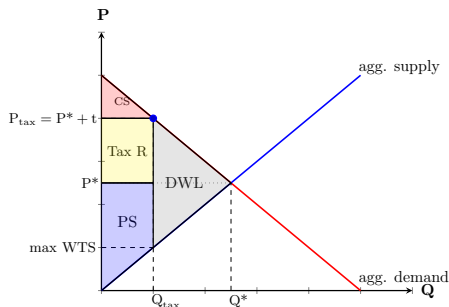


Market Failure: Taxation

- a large tax wedge creates more deadweight loss because it reduces the equilibrium quantities by more and thus excludes more producers and more consumers



small tax wedge, large base

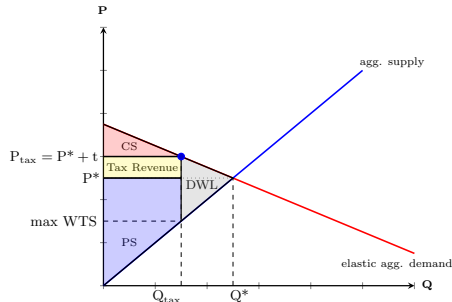
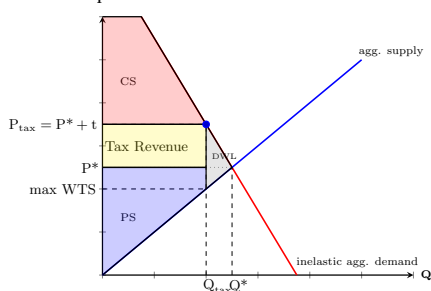


large tax wedge, small base

- tradeoff between wedge size and base size: tax revenue is roughly the same but the welfare effects are different

Market Failure: Taxation

- if goods are more elastic, a change in price leads to a much larger change in quantity and so a bigger deadweight loss

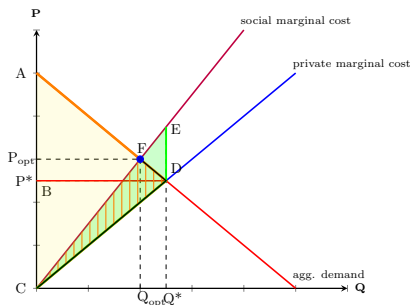


- also, notice how the tax incidence is different: the party with the more inelastic curve bears more of the DWL
 - when supply is less elastic than demand, producers give up more surplus
 - with a tax, producers with relatively inelastic supply are less affected by the change in prices and so can take on more of the tax incidence
 - when demand is less elastic than supply, consumers give up more surplus
 - consumers with relatively inelastic demand are less affected by the change in prices and so they end up taking up more of the tax incidence
 - (e.g. taxing cigarettes shifts most of the tax incidence onto the consumer $\Rightarrow$ while purchasing power has in general increased in France under Macron, the 2019 hike in sales tax on tobacco actually decreased purchasing power for the poorest 1% of the population who disproportionately smoke more)

Market Failure: Externalities

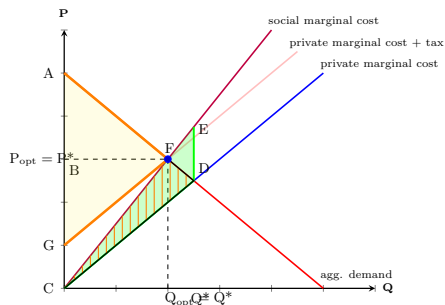
- an externality is a **cost** or a **benefit** of an economic transaction that affects a third party in such a way that the transactors do not choose the quantity of transaction while taking into account these third-party costs/benefits (= does not **internalize**)
- an externality can be **negative** or **positive**, of **production** or of **consumption**
 - a positive externality of production is such that the social benefit is higher than the private benefit = the good/service is **underproduced** or **underconsumed** compared to the social optimum
 - a negative externality of production is such that the social benefit is lower than the private benefit = the good/service is **overproduced** or **overconsumed** compared to the social optimum
- usually regulating externalities requires intervention that make producers (or consumers) internalize the externalities they impose on third parties
 - one method is a Pigovian tax, which is applied to goods that generate negative production externalities: tax is added to the marginal private cost to make producing at the socially optimal level also individually optimal
 - equivalently, impose quotas for production up to the socially optimal level
 - another method is to use a third party regulator that distributes rights to generate externalities in exchange for a payment that can be used to compensate affected third parties
 - all regulation methods require firms to disclose private information about costs, and any rescued surplus does not always benefit the third party

Market Failure: Externalities



welfare:

- $\Delta ADC = \Delta ABD + \Delta BDC = CS + PS = \text{total welfare (in absence of externalities)}$
- $\Delta ADC = \Delta ABD - \Delta CDE = CS + PS - \text{cost to third party} = \text{actual social welfare (in presence of externalities)}$
- $\Delta CDF = \text{welfare extracted from third parties that benefits producers and consumers but is cost to third-parties}$
- $\Delta DEF = \text{cost to third parties that is of no benefit to producers or consumers}$



welfare with Pigovian tax:

- new equilibrium at F is now socially optimal
- $\Delta AFG = \Delta AFB + \Delta BFG = CS + PS = \text{total welfare (with tax)}$
- $\Delta CFG = \text{surplus loss by producer after tax}$
- $\Delta CDF = \text{welfare returned to third parties that previously had been PS or CS}$
- $\Delta DEF = \text{welfare returned to third parties that previously had not belonged to either producer or consumer}$

Market Failure: Public Goods

- goods can be defined along two characteristics:
 - private goods are excludable (owner can exclude agents from its consumption) and rival (the consumption of the good by an agent makes it unavailable for consumption by other agents)
 - public goods tend to be non-excludable and non-rival
 - e.g. traffic signs, street lamps, national defense, roads
 - in reality, very few things are truly public goods, non-rivalry may give way to some amount of congestion
 - club goods (artificially scarce goods) tend to be non-rival but club rules exclude certain people from consumption (satellite TV, mobile network - both can saturate but at very high usage levels)
 - common resources tend to be rival but non-excludable (scarce natural resources are up for grabs, but using them means no one else can)
- all non-private goods of this sort are always underproduced or overconsumed
 - the positive externalities of their production is usually not considered in the actual production process because people don't internalize them naturally, those who do provide a free service to others (the free-riders) (think: planning on your economics notes jointly as a class but only half participate but everyone benefits)
 - the negative externalities of consumption of common goods is again usually not considered when we consume them and so we deplete resources faster than is socially optimal (think: climate change)

PS6 Exercise 2: Rivalry and Exclusion

For each of the following goods or bads, decide whether they are rival and whether they are excludable, and explain your answer. If you think the answer depends on factors not specified here, explain how.

	excludability	rivalry
a free public lecture at a university	non-excludable (public and free)	rival (limited capacity)
noise produced by aircraft around an airport	non-excludable (cannot stop people from hearing noise)	non-rival (one hearing the noise does not stop other from hearing the noise)
a public park	non-excludable (public and free)	rival (limited capacity)
a forest used by local people to collect firewood	non-excludable if access to forest is free	possibly non-rival if the number of wood collectors is too small relative to the size of the forest
seats in a theatre to watch a musical	excludable (need to pay)	rival (limited tickets, limited seats available)
bicycles available for hire to the public to travel around a city	excludable (not free)	rival (limited number of bikes + one person per bike)

Market Failure: Information Asymmetries

- one assumption of perfect competition is symmetrical information and perfect recall
 - in theory, this means everyone has access to all the information everyone else has access to, and thus both the characteristics and the action sets of a player are known by everyone
- in practice, information is often asymmetrical:
 - there are some pieces of information that aren't shared among the all players = **privately observed**
 - moral hazard = hidden action (player action is not perfectly observed, e.g. wage and effort)
 - adverse selection = hidden attributes (player characteristics are not perfectly observed, e.g. insurance)
- information asymmetry make competitive markets Pareto inefficient, as usually this leads to a poorly chosen tax/subsidy level or inability to set the right prices for different agents
 - market for lemons: two types of used cars (good peaches and bad lemons) sold by sellers who know the type and buyers who don't
 - if sellers sell to a buyer it must be that the buyer had a higher WTP than the seller had a willingness to sell
 - if perfect information: then buyers can distinguish between peaches and lemons $\Rightarrow$ two prices, one for each type + as long as there is trade, surplus is created
 - if asymmetric information: then there can only be one price since there is now only one type of good of unknown quality
 - buyers expect that half the cars are good and half the cars are bad, but because she cannot distinguish between the two, her WTP conditional on her own expectations is a lower value than in part (1); and because sellers know buyers will not have a WTP high enough to afford the good cars, so they all leave the market and what is left is a market of lemons and relative welfare loss

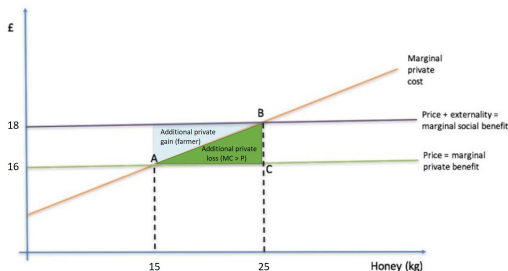
Problem Set 6: Exercise 1 - A Positive Externality

Imagine a beekeeper, who produces honey and sells it at a constant price per kilogram.

- 1 Draw a diagram with the quantity of honey on the horizontal axis, showing the marginal cost of honey production as an upward-sloping line, and the price of honey as a horizontal line. Find the amount of honey that the profit-maximizing beekeeper will produce.
- 2 For the beekeeper, the marginal private benefit (MPB: The benefit (in terms of profit, or utility) of producing or consuming an additional unit of a good for the individual who decides to produce or consume it, not taking into account any benefit received by others) of producing a kilo of honey is equal to the price. But since the bees benefit a neighbouring farmer, by helping to pollinate her crops, honey production has a positive external effect. Draw a line on your diagram to represent the marginal social benefit of honey production. Indicate the amount of honey that would be Pareto-efficient. Compare this quantity with the quantity chosen by the beekeeper.
- 3 Explain how the farmer and beekeeper could both be made better off through bargaining.

Problem Set 6: Exercise 1 - A Positive Externality

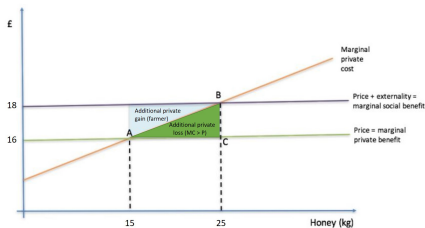
- 1 Draw a diagram with the quantity of honey on the horizontal axis, showing the marginal cost of honey production as an upward-sloping line, and the price of honey as a horizontal line. Find the amount of honey that the profit-maximizing beekeeper will produce.



- here, since he sells it a constant price per kilogram, the beekeeper (producer) sees the demand as a flat horizontal line
- then he produces at price (= marginal benefit) = MC, such that the equilibrium quantity is 15 and the price is 16.

Problem Set 6: Exercise 1 - A Positive Externality

- 2 For the beekeeper, the marginal private benefit is equal to the price. But since the bees benefit a neighbouring farmer, by helping to pollinate her crops, honey production has a positive external effect. Draw a line on your diagram to represent the marginal social benefit of honey production. Indicate the amount of honey that would be Pareto-efficient. Compare this quantity with the quantity chosen by the beekeeper.



- here, since the MSB is the MPB + the externality ($= 2$), so that is just another flat line above the demand curve by the size of the positive externality
- then, it is socially optimal for the beekeeper to produce at 25 (where $MSB = MC$), which is higher than the individually optimal quantity 15, and that is because the beekeeper does not internalise the positive externality when they produce
 - the new point 25 is Pareto efficient because at the social optimum producing any more would hurt the beekeeper via higher $MC > p$ (so not a Pareto improvement), but producing any less would hurt the farmer - since they don't get the full benefit of the positive externality (so not a Pareto improvement either)

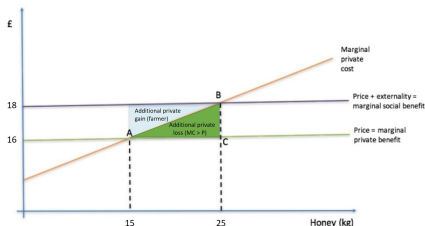
Problem Set 6: Exercise 1 - A Positive Externality

② cont'd

- then, it is socially optimal for the beekeeper to produce at 25 (where $MSB = MC$), which is higher than the individually optimal quantity 15, and that is because the beekeeper does not internalise the positive externality when they produce
 - the old point 15 is not clearly Pareto efficient or inefficient: the gain from moving from 15 to 25 for society is the sum of the blue and green triangles (it's the total social gain = extra production $\times$ marginal social benefit), and this outweighs the extra marginal cost incurred by the beekeeper (the green triangle).
 - in the presence of any compensatory/regulatory mechanism which takes the social gain and transfers part of it to the farmer to cover the cost, it can make both players better off = Pareto improvement possible from 15, so 15 not PE
 - if there is no compensatory mechanism then 15 is Pareto efficient since moving anywhere (including 25) is strictly harmful for the beekeeper (lower profits if they produce < 15 , marginal losses if they produce > 15)
 - it is NOT correct to say that 25 is not Pareto efficient because it hurts the beekeeper to get there from 15! (a point is PE if **from that point** there is no possible PI $\neq$ irrelevant to check if there was a PI in implementation of the PE outcome)
- notice that this assumes a constant externality per unit produced (which may not be)

Problem Set 6: Exercise 1 - A Positive Externality

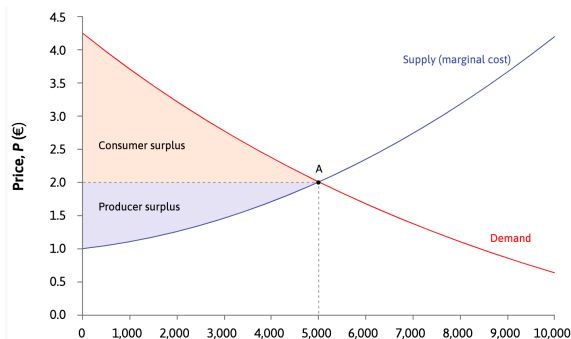
- 8 Explain how the farmer and beekeeper could both be made better off through bargaining.



- producing at 18 means that the farmer increases a cost equivalent to the green triangle (each extra unit incurs a marginal private cost), that is then suboptimal for him individually (he only gets the price on the green line but should produce above it to internalise the externality)
- so we'd need to subsidise the farmer's increased costs by transferring some of the benefit the farmer gets from this internalisation: that's the rectangle (sum of blue and green triangles), which is more than enough to cover the cost of the internalisation
- they could bargain and have the farmer pay a fixed sum equal to the green triangle + 1 dollar to the beekeeper: relative to the suboptimal non-internalised equilibrium, this is still a Pareto improvement for both players **which then allows the implementation of the PE outcome.**

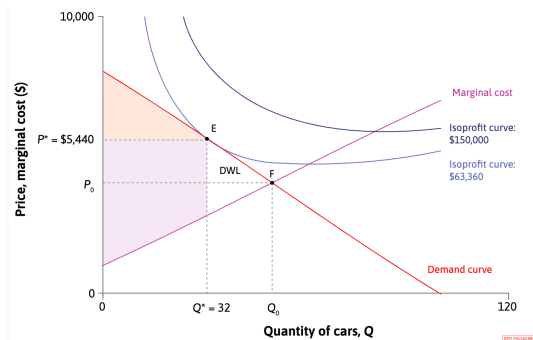
Problem Set 6: Reading

- ④ Initially, we assume that distributors and suppliers are price takers. The marginal cost of a cookie for suppliers is strictly increasing with quantities. The demand of the distributors is strictly decreasing in price.
- Graphically represent the supply and demand curves corresponding to this case. For simplicity, we will consider that the supply and demand curves are straight lines.
 - Next, plot the equilibrium price and quantities, as well as the distributor and supplier surpluses.



Problem Set 6: Reading

- ② It is now assumed that there is only one supplier who produces cookies. This supplier is therefore no longer a price taker.
- Knowing that the supplier is trying to maximize his profit, you will draw in a second graph the isoprofit curve of the supplier to find the equilibrium price and quantities. How do the equilibrium price and quantities differ from the case of perfect competition ?
 - From the graph, discuss the implications of this equilibrium in terms of supplier surplus, distributor surplus and total surplus

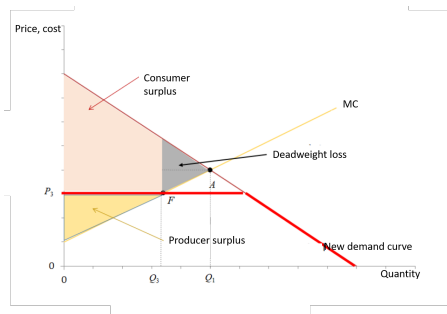


- with a monopoly, price is higher, quantity is lower; CS lower, PS ambiguous, DWL generated

Problem Set 6: Reading

8 Finally, we are interested in the case where the distributors are grouped together in a buying group. It is assumed that : the buying center sets a price above which it never buys the product and that the distributors seek to maximize their surplus.

- In a third graph, represent a supply curve (similar to that of the case of perfect competition), as well as the new demand curve.
- Compare the prices, quantities and surpluses between this third case and the first. In which case is the total surplus maximized ?
- Same question comparing the case with supplier market power and the case with a buying group.



- with a price ceiling: prices are artificially low compared to perfect competition, quantity also lower
- relative to perfect competition consumers protected (CS increases) at expense of producers (PS decreases), generates DWL
- relative to monopolies, it's clear CS is higher and PS is lower; no comment on DWL can be made (depends on elasticities)

Problem Set 6: Reading

- 4 In view of the preceding elements, under what conditions does the introduction of purchasing groups seem desirable to you ?
 - an introducing of purchasing groups can be interesting against a situation of monopoly if it is able to increase total surplus (this is easy if the DWL is too big in the case of a monopoly where market power is particular high)
 - but increasing total surplus may allow us to reach a PE outcome but the process might not be a PI (which makes it hard to implement)
- 5 What is one reason why introduction by a buying group might increase the wholesale price of a product ?
 - in the absence of a buying group, a producer/seller can practice price discrimination: selling to different buyers at different prices → this could be when you order in bulk and get a discount
 - in the presence of a buying group, everyone gets the same price and there is smaller margin of manoeuvre and negotiation
- 6 Under what condition does an increase in distributor surplus lead to an increase in consumer surplus ?
 - Here the distributor acts as an intermediary, so the only way an increase in their surplus can lead to an increase in final consumer surplus is if they **pass on the surplus gains**
 - So that distributors don't end up acting like monopolists in their submarket, we need to ensure healthy competition between distributors (which is not the case at the moment in Europe)

Next...

- **PS7 Exercise 2** due in class.
- **Quiz 4** next class, Chapters 5-6.
- Final exam on 6th of May.